

Lesson 1.13: Lab 2, Sheet Pan Nachos with a Label Discussion

Unit: Unit 1, Kitchen Skills and Smart Eating (Nutrition and Wellness) **Topic:** 1.3 Nutrients, MyPlate, and Labels **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 2 class periods of 40 minutes (Day 1 the Label Reading Quiz, prep, and label math; Day 2 the lab) **Room:** FACS lab with kitchen stations and an oven (a no-kitchen version is below)

Standards

- NYS module line: NYS N&W 1. Culinary Management b) Demonstrate safety and sanitation procedures when handling food and equipment
- NYS module line: NYS N&W 1. Culinary Management d) Prepare a simple recipe
- NYS module line: NYS N&W 1. Culinary Management g) Prepare and serve a simple meal
- NYS module line: NYS N&W 2. Nutrition Across the Lifespan a) Identify nutrient groups, common sources of nutrients, and nutrient requirements across the lifespan
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace
- CTE theme line: CTE FCL 4. Buying Goods and Services f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services
- NYS FACS Learning Standard 1 (Intermediate), Key Idea: Students will use an understanding of the elements of good nutrition to plan appropriate diets for themselves and others. They will know and use the appropriate tools and technologies for safe and healthy food preparation.; Performance Indicator A: Students understand the relationships among diet, health, and physical activities; evaluate their own eating patterns; and use appropriate technology and resources to make food selections and prepare simple, nutritious meals.; Performance Indicator B: Students apply principles of food safety and sanitation.
- CDOS Standard 3a Universal Foundation Skills: Interpersonal Skills
- National FCS 3.0: 14.3.3 Show the ability to select, store, prepare, and serve food that is nourishing and looks good on the plate.
- National FCS 3.0: 14.2.4 Examine sources of food and nutrition information, food labels included, for what they say about health.

(Four module lines is one more than the usual three. Lines 1. b), d), and g) are the lab; line 2. a) is the label task that runs while the pan bakes. The theme note from Lesson 1.11 applies: FCL 4. has no “labels” line, so f) is the informed-purchasing line and is quoted verbatim.)

Health class coordination

Health class typically covers: “choose healthy snacks” and sometimes a sodium or sugar limit as a number on a slide. Health does not cook.

This lesson goes further: students cook a real snack in a real kitchen, read the labels of the actual ingredients they used, compute the sodium in the plate in front of them, and judge it with a number. The label math is the piece no other class in the building does with a hot pan on the table.

Enduring understanding and essential question

Big idea: You can cook something fast, safe, and good, and still know exactly what is in it.

Reading the label of what you cooked is the honest version of reading a label. **Essential**

question: Is a snack you made yourself automatically healthier than one from a package?

Objectives

By the end of this lesson, students will be able to:

1. (Low: understand) Given the station card and their role card, state their own role's tasks in order and the two rules for the oven step (only the teacher opens the oven; the pan lands on a trivet and nobody touches it for one minute).
2. (Mid: apply) Working in a team of about seven in the eight course lab roles with the 40-minute flowchart, prepare and serve sheet pan nachos with a salsa bar, following safety and sanitation procedures, with every role completed and the station passing the end check.
3. (High: evaluate) Using the chip, cheese, bean, and corn labels, calculate the sodium in one serving of their own nachos, compare it to the daily value, and judge whether this is an “often” snack or a “sometimes” snack with the number as evidence.

Vocabulary

Tier 2 (general academic): flowchart, role, compute, judge Tier 3 (FACS): sheet pan, drain, layer, trivet, oven mitt, sodium, percent daily value. See the unit vocabulary list for the Spanish column.

Materials and setup

Per station (three stations of about seven students):

- 1 bag restaurant style tortilla chips, 13 ounces (a station of four or five gets half a bag)
- 1 cup (4 ounces) shredded Mexican blend cheese, pre-measured into a covered cup by the teacher or measured by the Prep Cook
- 1/2 can black beans (15.5 ounce can), can pre-opened by the teacher, drained and rinsed by the Prep Cook in a colander
- 1/3 can corn (15.25 ounce can), can pre-opened by the teacher, drained only



- 1 half sheet pan or quarter sheet pan, 1 sheet of heavy duty foil pre-cut to fit, 1 colander, 1 dry measuring cup (1 cup), 1 large spoon, 1 spatula, 2 oven mitts, 1 trivet, 7 paper plates, 7 forks, napkins, 1 sanitizing spray and 2 cloths
- Station Card, Role Cards, and Nacho Label Math handout: Handout 01.13 - Station Card Sheet Pan Nachos.md, Handout 01.13 - Role Cards Lab 2.md (the same eight roles as Lab 1: Station Manager, Head Cook, Second Cook, Prep Cook, Helper Cook, Toppings Chief, Plate Builder, Time Keeper), Handout 01.13 - Nacho Label Math.md
- Day 1: Assessment 01.3 - Label Reading Quiz.md, one per student, and a calculator per student
- The four empty packages (chip bag, cheese bag, bean can, corn can) with their Nutrition Facts panels, for the label task (the handout also prints the numbers)

Shared salsa bar, on a separate counter, set up before class and kept cold until the oven step:

- 1 jar salsa, 16 ounces, with a serving spoon
- 1 tub sour cream, 16 ounces, with a serving spoon
- Optional, no knives required: 1 can sliced black olives (drained), 1 bag pre-shredded lettuce.

Teacher station: oven preheated to 400 degrees F before students arrive, timer, oven mitts, clipboard with the lab rubric.

- Slides: Slides 01.13 - Lab 2 Sheet Pan Nachos with a Label Discussion.md (outline; build the deck from it; Day 1 and Day 2 are labeled)
- Setup notes: pre-cut foil the day before; pre-open one bean can and one corn can per station so no student touches a sharp lid; set the salsa bar before class; confirm the oven is at 400 before the bell; put oven mitts and a trivet at every station. Zero-knife lab: nothing is cut. Allergy check the day before: dairy (cheese, sour cream), and check the chip bag for shared-line warnings (some are made on lines with milk or wheat). A student who cannot eat dairy gets a foil-divided corner of the pan with no cheese and skips the sour cream.

Pre-assessment

The Kitchen Safety Exam (Topic 1.1) is the gate; nobody cooks without a pass on file. The Lab 1 rubric scores (Lesson 1.8) tell you which teams were slow on cleanup; give those teams the strongest Station Manager. The Lesson 1.11 Day 2 exit tickets and the Spot the Trick handouts tell you who can do percent daily value math; seat one strong label reader at each station as the Toppings Chief, who leads the label math. The Label Reading Quiz opens Day 1 of this lesson (the first 15 minutes, before the lab prep; see Assessment 01.3 - Label Reading Quiz.md), the same ticket-in pattern as the performance check on Lab 1 Day 1.

Do now and hook (Time: Day 1, 15 min; Day 2, 5 min)

Day 1, the do now is the Label Reading Quiz (Assessment 01.3 - Label Reading Quiz.md): calculator out, fifteen minutes, collected at minute 15. It is the Topic 1.3 grade and the ticket into the lab, the way the performance check opened Lab 1. A student who finishes early turns the quiz over and writes the nacho steps in order, as many as they can. Then the hook: show the 40-minute flowchart on the slide. “Here is the real order. Tomorrow you cook it. Today you learn your job, and you find out how much salt is in it before you eat it.”

Day 2, the do now is the setup routine: hair back, sleeves up, wash hands, apron on, role card in hand, station check against the station card list (everything on the counter, nothing missing). The Head Cook reads the card aloud and the Station Manager reports “station ready” to the teacher. The hook is the oven: “It is at 400. It is mine. Your pans go in when I say and come out when I say.”

Procedure

Day 1: the quiz, roles, and the label task (the quiz is the do now above)

Step	Teacher will	Students will	Time
1. Direct instruction: roles and the station card	Assign teams (about seven per station) and hand out the role cards, the same eight roles as Lab 1 with the nacho jobs on them: Station Manager, Head Cook, Second Cook, Prep Cook, Helper Cook, Toppings Chief, Plate Builder, Time Keeper (at seven students, the Station Manager keeps time). Walk the station card step by step on the slide, naming which role does each step. Say the two oven rules and have the class repeat them.	Read their role card. Highlight their steps on the station card. Say their role’s first task to a partner.	7 min
2. Guided discussion: the salsa	Show the salsa bar layout. Explain the	Plate Builders draw the traffic pattern	2 min

bar plan	cold rule from Lesson 1.2: salsa and sour cream stay in the refrigerator until the pans go in the oven, and the Plate Builder brings them to the bar then. Agree on the traffic pattern: one station at the bar at a time, Plate Builder serves, plates come to the bar, not the bar to the plates.	on the back of the station card. Everyone writes the cold rule under the salsa bar box.	
3. Learning activity: the Nacho Label Math	Hand out the four packages and the Nacho Label Math handout. Model the chips line on the board (serving size, sodium per serving, times the 13 servings the station uses). Assign one remaining ingredient per station: Station 1 cheese (4 servings), Station 2 beans (1.75), Station 3 corn (1). Each station's Toppings Chief reads that label aloud and the whole team computes its line; take the three numbers and write all four lines on the board. Then everyone: add the four to get the sodium in the whole pan; divide by 7 for	Read the station's label. Compute the station's line and report it. Copy the other three lines from the board. Compute the pan total, the per-person share, and the percent of a day. Answer the judgment question: often or sometimes, with the number.	10 min

	one person's share; add 2 tablespoons of salsa and 2 of sour cream; compare to the daily value of 2,300 mg. Circulate.		
4. Share and safety review	Take the per-person number from each station (they should land close to each other; about 680 mg with salsa and sour cream, about 30 percent of a day). Ask why the beans are the biggest surprise and what rinsing does (it removes roughly 40 percent of the sodium). Then the safety review: hot pan path from oven to trivet, mitts, no reaching over a pan, no student at the oven.	Toppings Chiefs give the number. Class votes often or sometimes. Time Keepers read the two oven rules aloud.	3 min

Day 1 timing: 15 (quiz) + 7 + 2 + 10 + 3 + 3 (closure) = 40 minutes.

Day 2: the lab

Step	Teacher will	Students will	Time
1. Line the pan and layer the chips (minutes 5 to 10)	Start the timer. Call "line and layer." Check each pan for even chips (one layer, not a mountain; the cheese has to reach the bottom).	Second Cook lines the pan with foil and spreads the chips in an even layer. Helper Cook brings the chips and cheese from the supply table. Toppings Chief measures the cheese. Prep Cook is at the sink.	5 min
2. Drain and top	Watch the sink:	Prep Cook drains	6 min

(minutes 10 to 16)	beans drained and rinsed in the colander, corn drained only. Check each pan before it moves: beans and corn spread, cheese over everything.	and rinses beans, drains corn. Second Cook spreads beans and corn; Toppings Chief adds the cheese edge to edge. Head Cook checks the pan against the station card and calls “pan ready.” Station Manager wipes spills; Helper Cook rinses the cans and recycles them.	
3. Oven, 400 degrees F, 8 to 10 minutes (minutes 16 to 26)	Load all pans with mitts. Set the timer for 8 minutes and check at 8; pull at 10 at the latest. During the bake, run the label discussion at the front: hold up the chip bag and the bean can, ask each station for its per-person sodium number from Day 1, ask which one ingredient they would swap to cut it most, and what that would do to the taste.	Plate Builder and Helper Cook bring salsa and sour cream from the refrigerator to the bar. Everyone else sits at home base for the label discussion. Toppings Chief speaks for the station. Time Keeper (at seven, the Station Manager) watches the clock and calls “two minutes.”	10 min
4. Remove and plate (minutes 26 to 32)	Pull each pan with mitts and set it on that station’s trivet. Say “one minute, hands off.” Then release stations to plate: spatula onto plates, one station at a time to the salsa bar.	Time Keeper counts the trivet minute. Head Cook portions with the spatula, about one seventh per plate. Plate Builder takes the station’s plates to the bar and serves salsa and sour cream (2 tablespoons each is	6 min

		a serving). Station Manager starts the wash water.	
5. Eat (minutes 32 to 36)	Sit with a plate. Ask the essential question at each station: is this healthier than the bag it came from? Score the rubric on the clipboard as you go (Objective 2 is scored here).	Eat. Answer the essential question at the table with the sodium number.	4 min

Questions to ask

Monitor understanding:

- What is your role's first step? What is your last one?
- Which ingredient has the most sodium per serving? Which has the most in the whole pan? Why are those different?
- Who opens the oven? Where does the pan go when it comes out?

Deepen learning:

- The beans are the healthiest thing on the pan and the saltiest. How can both be true?
- If you swapped the chips for a low sodium chip, how much would the per-person number drop? Is it worth it?
- Is a snack you made yourself automatically healthier than the bag? What decides that?

Check for understanding

Day 1, clipboard check during Step 3: for each station, yes or no on “reads serving size before sodium,” “multiplied by the station’s servings, not by servings per container,” and “per-person number is between 600 and 750 mg.” Day 2, the rubric on the clipboard: for each student, the role card tasks done, the two oven rules kept, and the station check at the end. A station that finishes the wash before the bell with the counter wiped and the pan clean passes the station check.

Closure (Time: Day 1, 3 min; Day 2, 4 min)

Day 1: exit line. Each student writes one line: “My role is . **My first step is** . The per-person sodium number is ___ mg.” Three read aloud. Sort by role: any role with two or more blank first steps gets a reminder on the Day 2 slide.

Day 2: clean up and station check (minutes 36 to 40). Station Manager runs the list on the station card by clean-up zone: pan washed and dried, foil in the trash, counter sprayed and

wiped, sink empty, mitts and trivet back, chairs in. Station Manager reports “station clean” to the teacher, who signs the station card. Exit line on the way out, spoken to the teacher: “Often or sometimes, and why?”

Differentiation and supports

- ELL: role cards have the tasks as numbered short lines with a picture in words for each; the station card’s steps are short numbered lines with the Tier 3 words in Vocabulary 01.md (a Spanish version of the steps and the two oven rules goes on the back of the card once the ENL teacher or a bilingual colleague has translated them); the oven rules are two pictures on the slide; pair each ELL student with a partner in a linked role (Second Cook with Prep Cook).
- IEP and 504 (general): a student who cannot stand for the whole period is the Toppings Chief or the Time Keeper at home base; the label math handout has the multiplication written as a fill-in with the servings already in the box; a calculator at every station; a student with a sensory issue around heat stays away from the trivet and plates last.
- Grade 6 support: the Nacho Label Math uses rounded numbers (beans as 2 servings, corn as 1) and asks for the pan total and the per-person share only; the percent of a day is given.
- Grade 8 stretch: compute the pan total for calories and saturated fat too, and rewrite the recipe with one swap (low sodium beans, reduced fat cheese, or half the chips) with the new per-person number; on Day 2, the Head Cook at a grade 8 station also plates one “MyPlate nacho plate” with a side of the salsa bar lettuce and reports whether it moved the plate closer to MyPlate.
- UDL checkpoint used: multiple means of action and expression (eight roles so every student has a physical task that fits them) and multiple means of representation (the flowchart as a slide, a station card, and a timer the teacher calls aloud).
- No-kitchen or no-machine alternative: Day 1 runs as written. Day 2 becomes “build a nacho plate”: each team gets the Nutrient Sort Cards plus six nacho ingredient cards (chips, cheese, beans, corn, salsa, sour cream) with the label numbers on the back. Teams build the plate, compute the sodium for the plate exactly as on Day 1, then rebuild it with two swaps to bring the sodium under 400 mg per person while keeping at least three MyPlate groups on the plate. The Station Manager’s job becomes the recorder. If a microwave is available, a single microwave version works: chips, beans, corn, and cheese on a paper plate for 45 seconds, one plate per team, teacher at the microwave.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. State own role tasks and the two oven rules	N&W 1. b); HSW 4. c); Std 1 PI B	Role cards and station card, Day 1 Step 1; exit line	Formative; exit line checked; summative on the Lab 2 rubric,

			row 1
2. Prepare and serve nachos safely in a team, station passes the end check	N&W 1. b), d), g); HSW 4. c), e); Std 1 PI B; CDOS 3a Interpersonal Skills; National FCS 14.3.3	The lab, Day 2 Steps 1 to 5 and closure	Summative; Lab 2 rubric, rows 2 and 4, scored on the clipboard during the lab
3. Compute the sodium in one serving and judge the snack with evidence	N&W 2. a); FCL 4. f); Std 1 PI A; National FCS 14.2.4	Nacho Label Math, Day 1 Step 3; label discussion, Day 2 Step 3	Summative; Lab 2 rubric, row 3, scored from the handout and the Day 2 discussion

Lab 2 rubric (the lab rubric pattern from the grading plan: three objectives plus a safety and sanitation row, each 1 to 4, 16 points; counts in Labs and Projects)

Criterion	Not Yet 1	Approaching 2	Meets 3	Exceeds 4
1. Role and oven rules	Could not state own role's steps or an oven rule; needed the card read aloud	Stated the role's steps out of order or one oven rule	Stated own role's steps in order and both oven rules	Stated own role and a teammate's role, both oven rules, and why each rule exists
2. Role tasks and teamwork	Skipped role tasks or left them for a teammate to do, or the station was not reset on time	Did most role tasks; needed one reminder to do a task or to pass a teammate what they needed; station reset late	Did every role task on time, passed teammates what they needed without being asked, station reset on time	Did every role task, helped another role without being asked, station reset early
3. Label math and judgment	Table incomplete or the per-person number missing	Table complete but one computation wrong, or the judgment has no number	Table complete, per-person number within range, judgment states often or sometimes with the number	All of Meets plus a swap that cuts the sodium, with the new number computed
4. Safety and sanitation (every lab, same row all year)	A safety violation: hands not washed, hair loose over food, bare hand	Hands, hair, and apron done, but one reminder needed during	Washed hands at the start; hair and apron done; mitts and trivet used for	Meets, and called out a hazard for the team before the teacher saw it,

	on a hot pan, a student at the oven, food dropped and served, or a station left dirty after the check	the lab (a towel near the oven, reaching over a pan, a wet floor left)	every hot pan; station passed the check the first time	or fixed one at another station without being asked
Total				/16

How to use: score rows 2 and 4 on the clipboard during Day 2 (row 2 is the role tasks and teamwork; every safety and sanitation step is scored on row 4 only, so a missed step costs a student once); score rows 1 and 3 from the Day 1 exit line and the handout after class. Write one line in the notes box for the next lab (what this student should try next). A safety violation is a 1 on row 4 and a zero for that lab's participation, per the grading plan, and a conversation.

Homework

None. Optional: read the label on one snack at home and compute the sodium in your usual portion, not the serving. Bring the number.

Connections

Inside the building: the math teacher (the same ratio work as Lesson 1.11, now with a pan on the table) and the cafeteria manager, who can show the sodium number on the school lunch menu if the district posts one. FACS gives the cafeteria a class that will read the posted numbers; the cafeteria gives FACS a real menu for the \$40 project comparison in Topic 1.4.

Sources

- NYS Middle Level CTE FACS Content Module 05, Nutrition and Wellness (nyc-tecenter.org, June 2018), lines 1. b), 1. d), 1. g), and 2. a). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 04, Health, Safety, and Wellness, lines 4. c) and 4. e); Theme Module 03, Financial and Consumer Literacy, line 4. f). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 1, Intermediate, FACS strand. Wording per 02 Standards/STANDARDS SOURCES - web research.md, section 2.
- CDOS Standard 3a, Interpersonal Skills. Same file.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 14.3.3 and 14.2.4.
- Sodium daily value (2,300 mg) and the 5 and 20 rule: U.S. Food and Drug Administration, "How to Understand and Use the Nutrition Facts Label" (fda.gov). Rinsing canned beans removes about 41 percent of the sodium: Duyff, Jones, and

Mount, “Sodium reduction in canned beans after draining, rinsing,” *Journal of Culinary Science and Technology*, 2011.

- My own materials: my Sheet Pan Nachos with Salsa Bar teacher sheet (the 40-minute flowchart, ingredient amounts per station, prep notes, the zero-knife policy, and the “Nutrition Facts discussion while it bakes” idea). Rewritten here for three stations of seven with the teacher on oven duty. The label numbers on the handouts are typical values for store brand products and must be replaced with the numbers on the packages you actually buy.

Teacher notes

- If Day 1 runs short, cut Step 2 (the salsa bar plan) to one minute and let the Plate Builders figure out the traffic pattern on Day 2 with you at the bar. Never cut the quiz or Step 3; the quiz is the Topic 1.3 grade and the label math is Objective 3 and the rubric row.
- If Day 2 runs short, the bake is fixed at 8 to 10 minutes; the only place to save time is plating. Have the Plate Builders line up the plates during the bake.
- Per-station cost, estimated at Long Island supermarket store brand prices, September 2026: chips 13 oz \$3.99; cheese 8 oz \$3.49 (half used; buy one bag for two stations, \$1.75); black beans \$1.29 (half used; one can for two stations, \$0.65); corn \$1.19; salsa 16 oz \$3.29 shared by three stations, \$1.10; sour cream 16 oz \$2.49 shared, \$0.83; foil, about \$0.30. About \$9.80 per station buying shared items across the class, or about \$12.20 if every station gets its own cheese bag and bean can. Optional olives and lettuce add about \$1.35 per station. A class of three stations runs about \$30 to \$41, well under the \$20 per station line.
- Common mistake: a pan with the chips piled in a mound, so the middle stays cold and the cheese never reaches the bottom. Say “one layer, you should see foil between some chips” at Step 1.
- Second common mistake: the Prep Cook rinses the corn too and it goes watery, or does not rinse the beans and the pan tastes of the can. Beans: drain and rinse. Corn: drain only. It is on the station card twice.
- The teacher at the oven is not a suggestion in this version. My original sheet had students rotate mitts and pull their own pans; with three stations of seven and grade 6 in the room, the teacher pulls.
- The label discussion during the bake is the part that makes this a nutrition lesson and not just a snack. Do not skip it to “let them relax.” Ten minutes is enough for the number, the swap question, and the beans surprise.
- Keep the four empty packages. They are the review set for anyone retaking the Label Reading Quiz and one of the stations in Lesson 1.16 (unit price).